

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-1:Lecture-9
Boolean Algebra

CONTENTS

Module-1 (Part-2)

- ❖ Logic gates and universal gates
- ❖ Tri-state logic
- ❖ **Boolean algebra**
- ❖ SOP and POS minimization
- ❖ K-maps upto 4-variables,
- ❖ Don't care logic

CONTENTS

Lecture-9

- ❖ Boolean Algebra (cont...)
- ❖ Gate Level Implementation

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Boolean Algebra

BOOLEAN ALGEBRA

T1 : Properties of 0

(a) $0 + A = A$

(b) $0 \cdot A = 0$

T2 : Properties of 1

(a) $1 + A = 1$

(b) $1 \cdot A = A$

T3 : Commutative Law

(a) $A + B = B + A$

(b) $A B = B A$

T4 : Associate Law

(a) $(A + B) + C = A + (B + C)$

(b) $(A B) C = A (B C)$

T5 : Distributive Law

(a) $A (B + C) = A B + A C$

(b) $A + (B C) = (A + B) (A + C)$

(c) $A + A'B = A + B$

T6 : Indempotence (Identity) Law

(a) $A + A = A$

(b) $A A = A$

T7 : Absorption (Redundance) Law

(a) $A + A B = A$

(b) $A (A + B) = A$

T8 : Complementary Law

(a) $X + X' = 1$

(b) $X \cdot X' = 0$

T9 : Involution

(a) $X'' = X$

T10 : De Morgan's Theorem

(a) $(X + Y)' = X' \cdot Y'$

(b) $(X \cdot Y)' = X' + Y'$

THEOREMS OF BOOLEAN ALGEBRA

P1: $x \cdot \bar{x} = 0$

P2: $x + \bar{x} = 1$

P3: $x \cdot 1 = x$

P4: $x + 0 = x$

P5: $x + y = y + x$

P6: $x \cdot y = y \cdot x$

P7: $x + (y + z) = (x + y) + z$

P8: $x \cdot (y \cdot z) = (x \cdot y) \cdot z$

P9: $x(y + z) = xy + xz$

P10: $x + yz = (x + y)(x + z)$

T1: $x \cdot x = x$

T2: $x + x = x$

T3: $x \cdot 0 = 0$

T4: $x + 1 = 1$

T5: $x + xy = x$

T6: $x(x + y) = x$

T7: $\overline{x + y} = \bar{x} \cdot \bar{y}$

T8: $\overline{xy} = \bar{x} + \bar{y}$

T9: $(x')' = x$

T10: $xy + \bar{x}z + yz = xy + \bar{x}z$

T11: $(x + y)(\bar{x} + z)(y + z) = (x + y)(\bar{x} + z)$

BOOLEAN ALGEBRA

Algebraic Method

T1: $x \cdot x = x$

Proof:

L.H.S.	$= x \cdot x$	
	$= x \cdot x + 0$	----- P4
	$= x \cdot x + x \cdot x'$	----- P1
	$= x \cdot (x + x')$	----- P9
	$= x \cdot 1$	----- P2
	$= x =$ R.H.S.	----- P3

BOOLEAN ALGEBRA

Algebraic Method

T2: $x + x = x$ (Apply Duality)

Proof:

$$\text{L.H.S.} = x + x$$

$$= (x + x).1 \quad \text{----- P3}$$

$$= (x + x).(x + x') \quad \text{----- P2}$$

$$= x + x.x' \quad \text{----- P10}$$

$$= x + 0 \quad \text{----- P1}$$

$$= x = \text{R.H.S.} \quad \text{----- P4}$$

BOOLEAN ALGEBRA

Algebraic Method

T3: $x + 1 = 1$

Proof:

$$\text{L.H.S.} = x + 1$$

$$= (x + 1).1 \quad \text{----- P3}$$

$$= (x + 1).(x + x') \quad \text{----- P2}$$

$$= x + 1.x' \quad \text{----- P10}$$

$$= x + x'.1 \quad \text{----- P6}$$

$$= x + x' \quad \text{----- P3}$$

$$= 1 = \text{R.H.S.} \quad \text{----- P2}$$

BOOLEAN ALGEBRA

Algebraic Method

T4: $x \cdot 0 = 0$ (Apply Duality)

Proof:

L.H.S.	$= x \cdot 0$	
	$= x \cdot 0 + 0$	----- P4
	$= x \cdot 0 + x \cdot x'$	----- P1
	$= x \cdot (0 + x')$	----- P9
	$= x \cdot (x' + 0)$	----- P5
	$= x \cdot x'$	----- P4
	$= 0 =$	R.H.S. ----- P1

BOOLEAN ALGEBRA

Algebraic Method

T5: $x + xy = x$

Proof:

L.H.S. = $x + xy$	
$= x.1 + xy$	----- P3
$= x.(1 + y)$	----- P9
$= x.(y + 1)$	----- P5
$= x.1$	----- T4
$= x = \text{R.H.S.}$	----- P3

BOOLEAN ALGEBRA

Algebraic Method

T6: $x(x + y) = x$ (Apply Duality)

Proof:

L.H.S. = $x(x + y)$	
$= (x + 0)(x + y)$	----- P4
$= x + 0.y$	----- P10
$= x + y.0$	----- P6
$= x + 0$	----- T3
$= x = \text{R.H.S.}$	----- P4

BOOLEAN ALGEBRA

Algebraic Method

T10: $xy + x'z + yz = xy + x'z$

Proof:

$$\begin{aligned}\text{L.H.S.} &= xy + x'z + yz \\ &= xy + x'z + yz \cdot 1 \\ &= xy + x'z + yz(x + x') \\ &= xy + x'z + yzx + yzx' \\ &= xy + xyz + x'z + x'yz \\ &= xy(1 + z) + x'z(1 + y) \\ &= xy + x'z = \text{R.H.S.}\end{aligned}$$

BOOLEAN ALGEBRA

✓ The **operator precedence** for evaluating Boolean Expression is

- Parenthesis
- NOT operation
- AND Operation
- OR operation

✓ The function below has **8 literals** (number of letters) and **2 Boolean variables** x and y

$$F = xy' + x'y + xy + (xy)'$$

✓ **Note:** When a Boolean Expression is implemented, each variable in the expression is called as ***literals***.

BOOLEAN ALGEBRA

Algebraic Method

T11: $(x + y)(x' + z)(y + z) = (x + y)(x' + z)$ (Apply Duality)

Proof: L.H.S. $= (x + y)(x' + z)(y + z)$

$$= (xx' + xz + yx' + yz)(y + z)$$

$$= (xz + yx' + yz)(y + z)$$

$$= xzy + xzz + yx'y + yx'z + yzy + yzz$$

$$= xyz + xz + x'y + x'yz + yz + yz$$

$$= xyz + xz + x'y + x'yz + yz$$

$$= xz(y + 1) + x'y(1 + z) + yz$$

$$= xz + x'y + yz$$

$$= xz + x'y + yz + xx'$$

$$= xz + yz + xx' + yx'$$

$$= (x + y)z + (x + y)x'$$

Let $(x + y) = p$

$$= pz + px'$$

$$= p(z + x')$$

$$= (x + y)(x' + z) = \text{R.H.S.}$$

BOOLEAN ALGEBRA

Examples: Solve the following

(i) $F = xy' + x'y + xy + (xy)'$

(ii) $F = x[y + z(xy + xz)']$

Solution: (i) $F = xy' + x'y + xy + (xy)'$
 $= xy' + x'y + xy + x' + y'$
 $= xy' + y' + x'y + x' + xy$
 $= (x + 1)y' + x'(y + 1) + xy$
 $= y' + x' + xy$

$$= x' + y' + xy$$

$$= (xy)' + xy$$

Let $(x + y) = p$

$$= p' + p = 1$$

BOOLEAN ALGEBRA

Solution: (ii) $F = x[y + z(xy + xz)']$

$$= xy + xz(xy + xz)'$$

$$= xy + xz(x' + y')(x' + z')$$

$$= xy + xz(x'x' + x'z' + y'x' + y'z')$$

$$= xy + xz(x' + x'z' + x'y' + y'z')$$

$$= xy + xz(x'(1 + z' + y') + y'z')$$

$$= xy + xz(x' + y'z')$$

$$= xy + xzx' + xzy'z' = xy \quad \because xx' = 0 \text{ \& } zz' = 0$$

BOOLEAN ALGEBRA

Examples: Solve $[(yz' + x'u)(xy' + zu')]'$ to minimum number of literals

Solution:

$$\begin{aligned} & [(yz' + x'u)(xy' + zu')]' \\ &= [yz'xy' + yz'zu' + x'uxy' + x'uzu']' \\ &= [0 + 0 + 0 + 0]' \\ &= [0]' = 1 \end{aligned}$$

ASSIGNMENT QUESTIONS

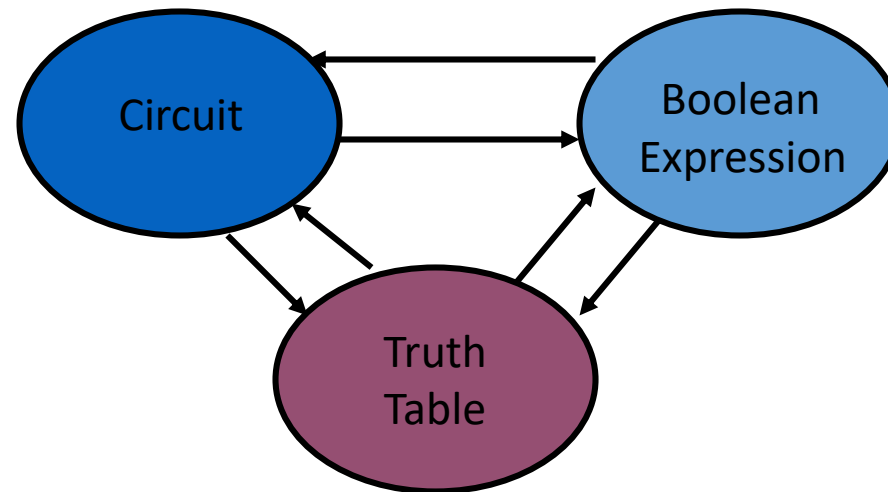
Question: Prove the following

(i) $x + x'y = x + y$

(ii) $xy + \overline{xz} + x\overline{y}z(xy + z) = 1$

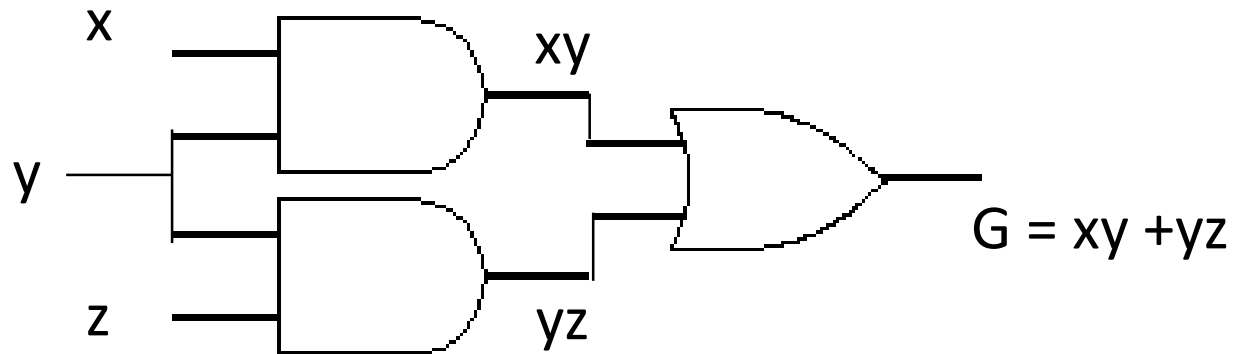
GATE LEVEL IMPLEMENTATION

- ✓ Need to transition between Boolean expression, truth table, and circuit (symbols).
- ✓ Converting between truth table and expression is easy.
- ✓ Converting between expression and circuit is easy.
- ✓ More difficult to convert to truth table.



GATE LEVEL IMPLEMENTATION

- ✓ We will learn how to transition between equation, symbols, and truth table.

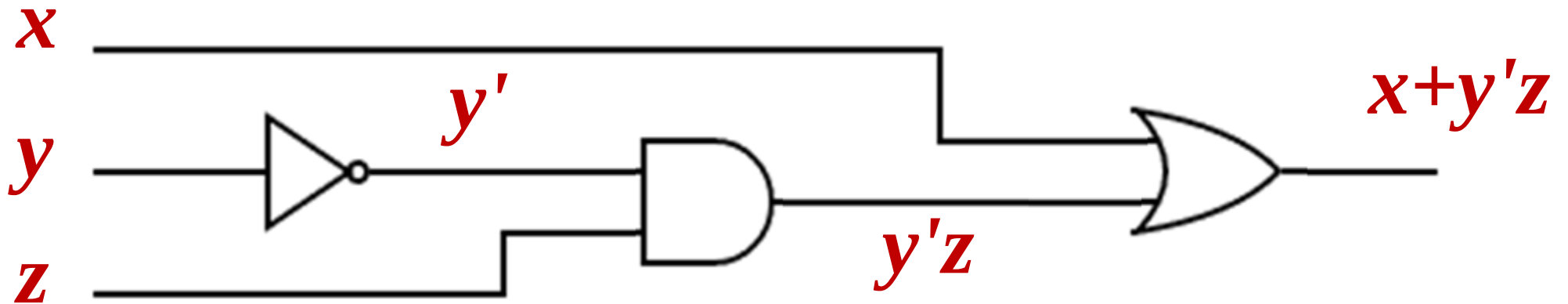


x	y	z	xy	yz	G
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	1	1

GATE LEVEL IMPLEMENTATION

- ✓ Consider a Boolean Function

$$F = x + y'z$$



GATE LEVEL IMPLEMENTATION

Need of Reduction of Algebraic Function

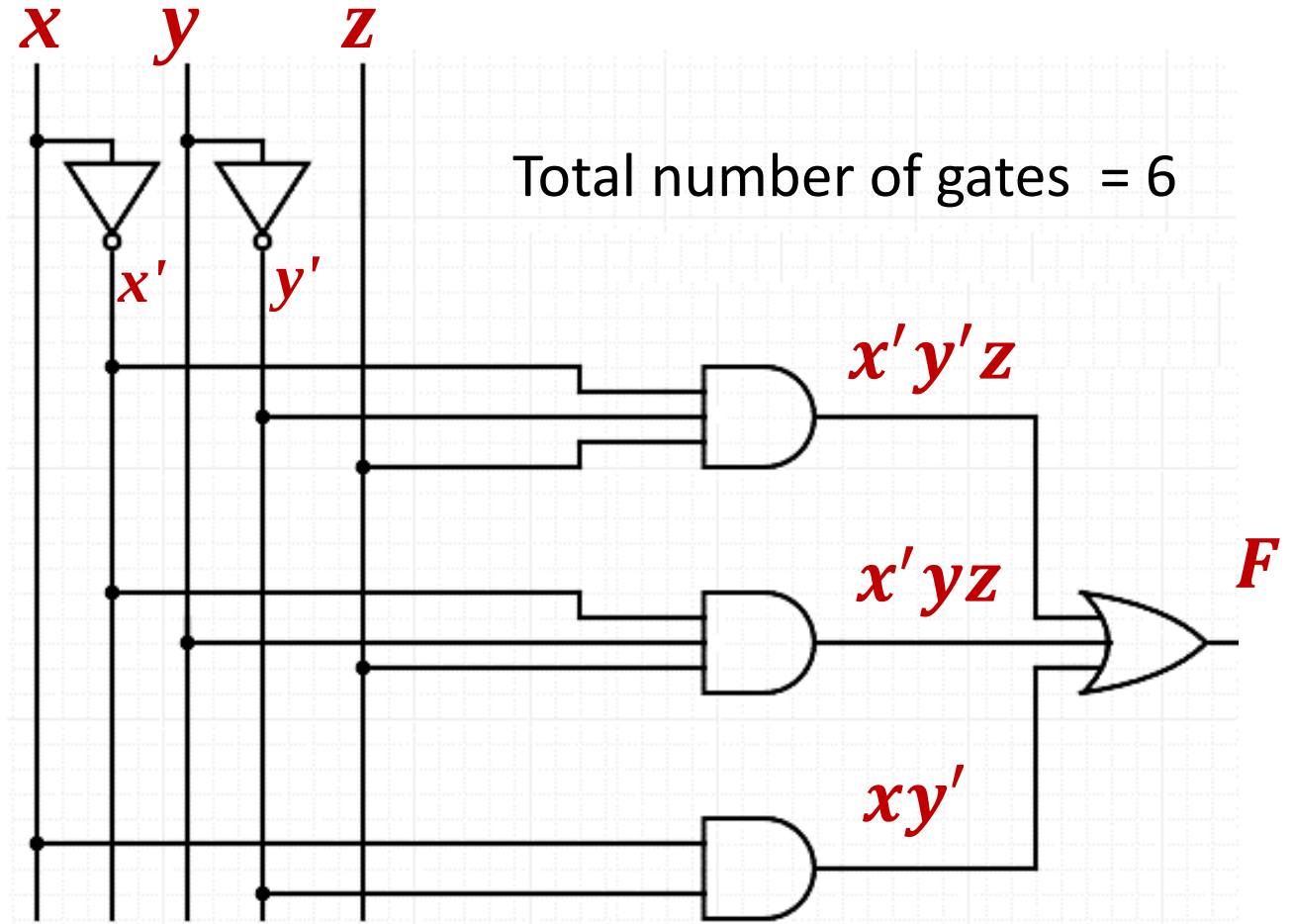
- ✓ Advantage is that the number of gates required is reduced in order to implement a Boolean function.

- ✓ Consider a Boolean function

$$F = x'y'z + x'yz + xy'$$

- ✓ After reduction F becomes

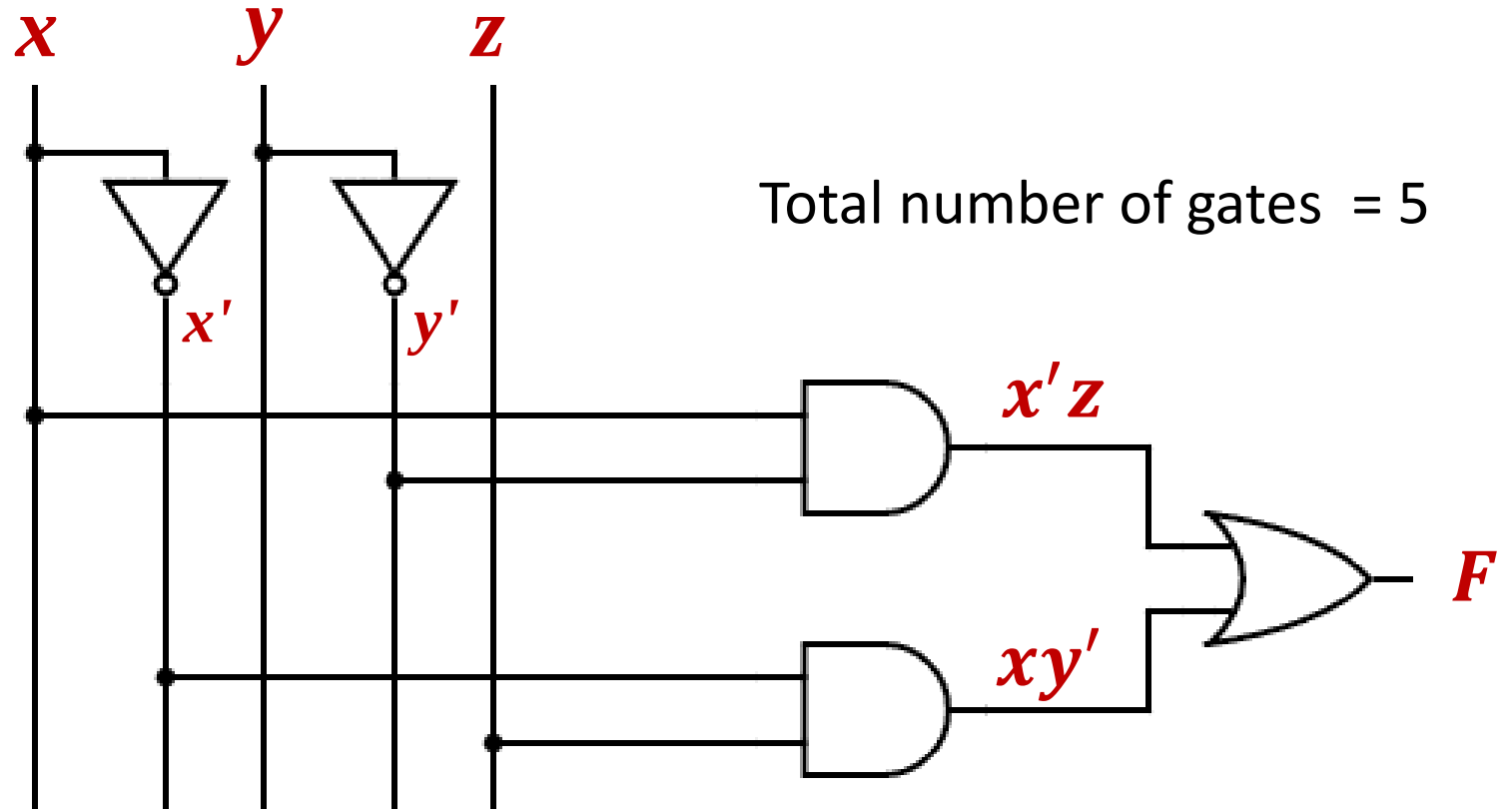
$$F = xy' + x'z$$



GATE LEVEL IMPLEMENTATION

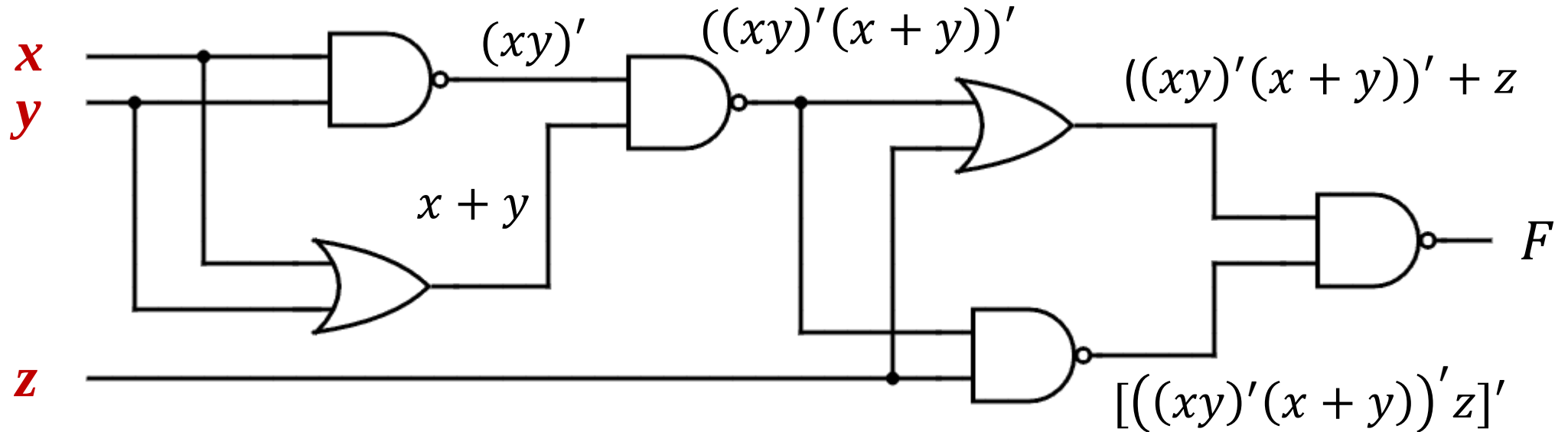
Need of Reduction of Algebraic Function

$$F = xy' + x'z$$



GATE LEVEL IMPLEMENTATION

- ✓ Find the Boolean Expression F for the following circuit



$$F = ((xy)'(x + y))' + z)[((xy)'(x + y))'z]'$$

Thank You